



Daffodil
International
University

Department of Computer Science And Engineering (CSE)

Faculty of Science and Information Technology

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ASSIGNMENT - 01

Course Title : Instrumentation and Control
Course Code : CSE325
Assignment On : Signal Flow Graph

05

Submitted By	Submitted To
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Date of Submission: 18-07-2026

Comment by Course Teacher:

good

Signal Flow Graph (SFG)

1.40 Draw the signal flow graph for following set of equations.

$$x_2 = x_1 + ax_5$$

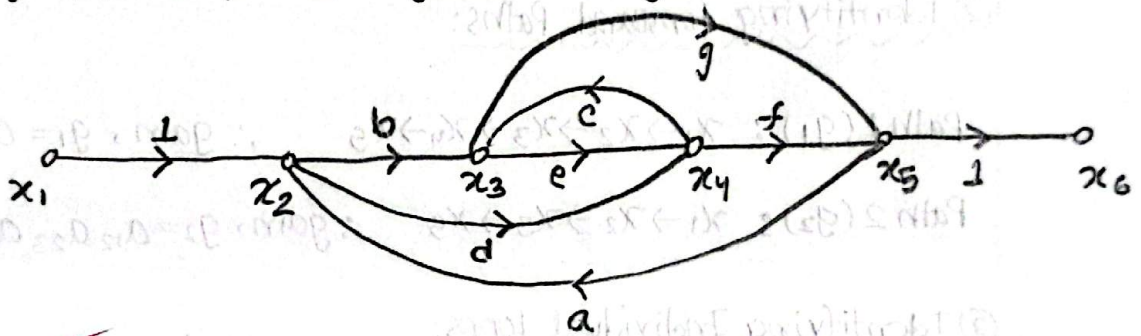
$$x_3 = bx_2 + cx_4$$

$$x_4 = dx_2 + ex_3$$

$$x_5 = fx_4 + gx_3$$

$$x_6 = x_5$$

Soln: The required signal flow graph is -



1.41 For the system represented by the given equations find the transfer function x_5/x_1 by the help of signal flow graph technique

$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$

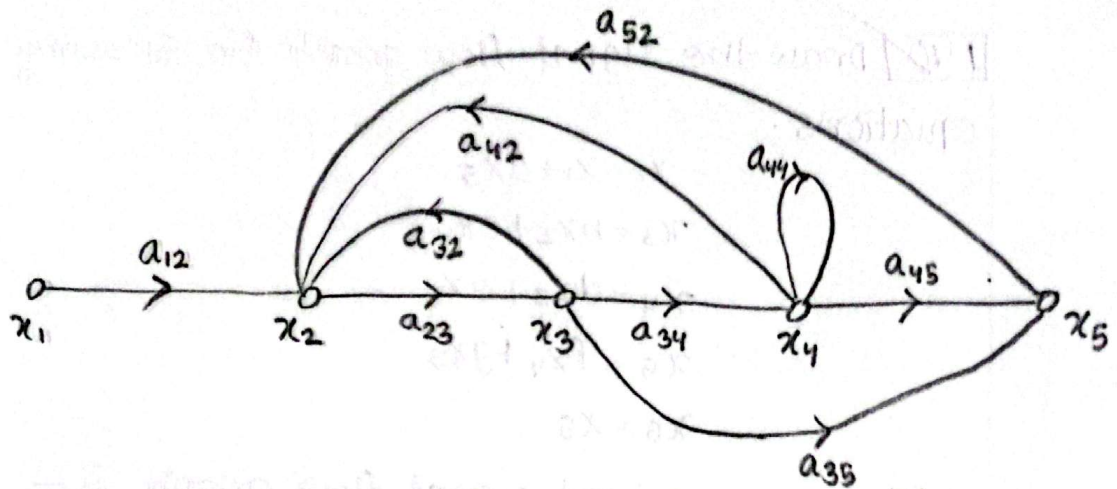
$$x_3 = a_{23}x_2$$

$$x_4 = a_{34}x_3 + a_{44}x_4$$

$$x_5 = a_{35}x_3 + a_{45}x_4$$

where x_1 is the input variable and x_5 is the output variable.

Soln ① Drawing signal flow Graph:



② Identifying Forward Paths:

Path 1 (g_1): $x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow x_5$ $\therefore$ gain, $g_1 = a_{12} a_{23} a_{34} a_{45}$

Path 2 (g_2): $x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_5$ $\therefore$ gain, $g_2 = a_{12} a_{23} a_{35}$

③ Identifying Individual Loops:

Loop 1 (L_1): $x_2 \rightarrow x_3 \rightarrow x_2$ $\therefore$ Gain, $L_1 = a_{23} a_{32}$

Loop 2 (L_2): $x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow x_2$ $\therefore$ Gain, $L_2 = a_{23} a_{34} a_{42}$

Loop 3 (L_3): $x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow x_5 \rightarrow x_2$ $\therefore$ Gain, $L_3 = a_{23} a_{34} a_{45} a_{52}$

Loop 4 (L_4): $x_2 \rightarrow x_3 \rightarrow x_5 \rightarrow x_2$ $\therefore$ Gain, $L_4 = a_{23} a_{35} a_{52}$

Loop 5 (L_5): self loop at x_4 $\therefore$ Gain $L_5 = a_{44}$

④ Identifying Non-touching Loops:

product of two non-touching loops,

$$L_1 L_5 = a_{23} a_{32} a_{44}$$

$$L_4 L_5 = a_{23} a_{35} a_{44}$$

⑤ Calculation of Δ and Δ_k :

According to Mason's Gain formula:

$$\Delta = 1 - \sum (\text{all individual loops}) + \sum (\text{product of two non touching loops})$$

$$\therefore \Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + (L_1 L_5 + L_4 L_5)$$

$$= 1 - (a_{23}a_{32} + a_{23}a_{34}a_{45}a_{54} + a_{23}a_{34}a_{45}a_{52} + a_{23}a_{35}a_{52} + a_{44}) + (a_{23}a_{32}a_{44} + a_{23}a_{35}a_{44})$$

$$\therefore \Delta_1 = 1 - \sum \text{single loops not touching path } g_1 = 1 - 0 = 1$$

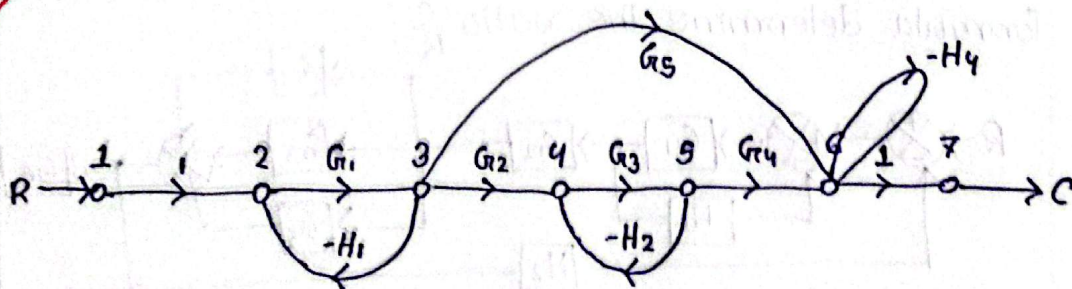
$$\therefore \Delta_2 = 1 - \sum \text{single loops not touching path } g_2 = 1 - L_5 = 1 - a_{44}$$

$$\therefore \text{Transfer function, } T = \frac{X_5}{X_1} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$= \frac{a_{12}a_{23}a_{34}a_{45} \cancel{1} + a_{12}a_{23}a_{35}(1 - a_{44})}{1 - (a_{23}a_{32} + a_{23}a_{34}a_{44} + a_{23}a_{34}a_{45}a_{52} + a_{23}a_{35}a_{52} + a_{44}) + (a_{23}a_{32}a_{44} + a_{23}a_{35}a_{44})}$$

(Ans)

✓ 1.42 For the given signal flow graph find the ratio C/R



Soln:

The gain of forward paths $g_1 = \pm G_1 G_2 G_3 G_4$

$$g_2 = G_1 G_5$$

Individual loops: $L_1 = -G_1 H_1$

$$L_2 = -G_3 H_2$$

$$L_3 = -G_1 G_2 G_3 H_3$$

$$L_4 = -H_4$$

Two non-touching loops: $L_1 L_2 = G_1 H_1 G_3 H_2$

$$L_1 L_4 = G_1 H_1 H_4$$

$$L_2 L_4 = G_3 H_2 H_4$$

$$L_3 L_4 = G_1 G_2 G_3 H_3 H_4$$

Three non touching loops: $L_1 L_2 L_4 = -G_1 H_1 G_3 H_2 H_4$

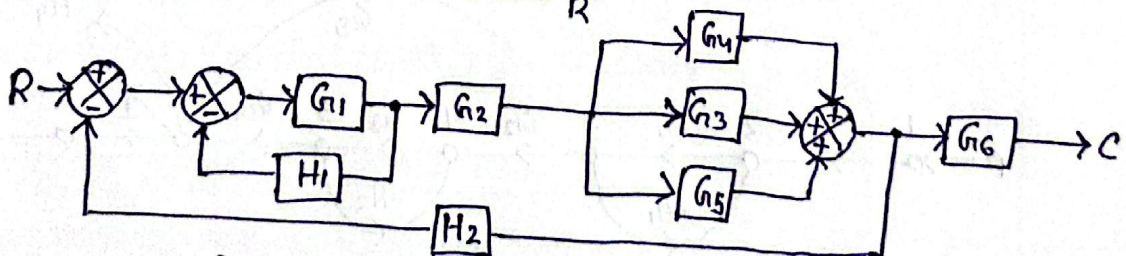
$$\therefore \Delta_1 = 1 - 0 = 1 \quad \therefore \Delta_2 = 1 + G_3 H_2$$

$$\therefore \Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1 L_2 + L_1 L_4 + L_2 L_4 + L_3 L_4) - (L_1 L_2 L_4)$$

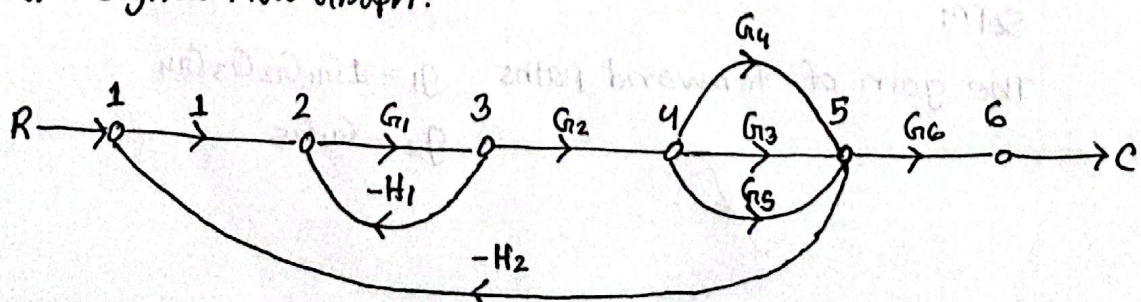
$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 G_4 + G_1 G_5 (1 + G_3 H_2)}{1 + G_1 H_1 + G_3 H_2 + G_1 G_2 G_3 H_3 + H_4 + G_1 G_3 H_1 H_2 + G_1 G_5 H_1 H_4 + G_3 H_2 H_4 + G_1 G_2 G_3 H_3 H_4 + G_1 G_3 H_1 H_3 H_4}$$

(Ans)

1.43 Obtain signal flow graph representation for a system whose block diagram is given below and using Mason's gain formula determine the ratio $\frac{C}{R}$



Soln: Signal Flow Graph:



The gain of forward paths: $g_1 = G_1 G_2 G_3 G_6$

$$g_2 = G_1 G_2 G_4 G_6$$

$$g_3 = G_1 G_2 G_5 G_6$$

Individual Loops: $L_1 = -G_1 H_1$

$$L_2 = -G_1 G_2 G_3 H_2$$

$$L_3 = -G_1 G_2 G_4 H_2$$

$$L_4 = -G_1 G_2 G_5 H_2$$

Two non-touching loops: none

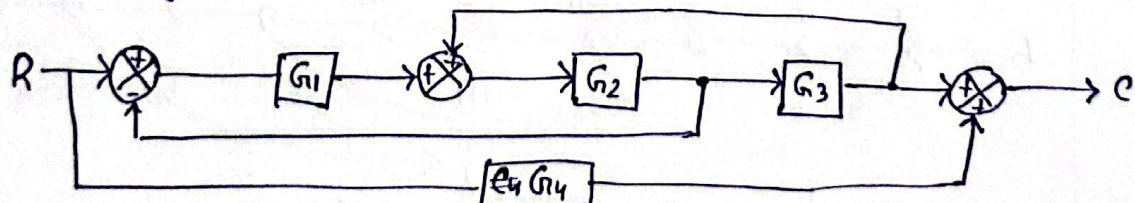
Calculating Δ and Δ_k : $\Delta_1 = 1 - 0 = 1$, $\Delta_2 = 1 - 0 = 1$, $\Delta_3 = 1 - 0 = 1$

$$\therefore \Delta = 1 - (L_1 + L_2 + L_3 + L_4)$$

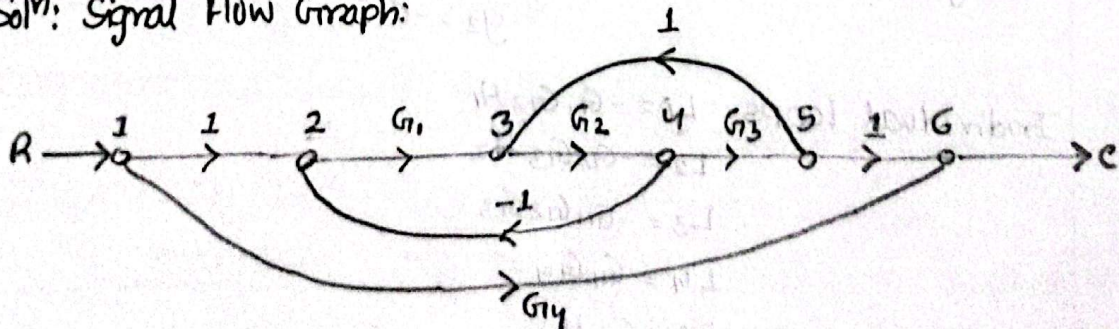
$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 G_6 + G_1 G_2 G_4 G_6 + G_1 G_2 G_5 G_6}{1 + G_1 H_1 + G_1 G_2 G_3 H_2 + G_1 G_2 G_4 H_2 + G_1 G_2 G_5 H_2}$$

(Ans)

1.44 Obtain the transfer function for $\frac{C(s)}{R(s)}$ given block diagram using block diagram reduction technique or signal flow graph (Mason's gain formula).



Soln: Signal Flow Graph:



Gain of forward paths: $g_1 = G_1 G_2 G_3$

$$g_2 = G_4$$

Individual loops: $L_1 = -G_1 G_2$

$$L_2 = G_2 G_3$$

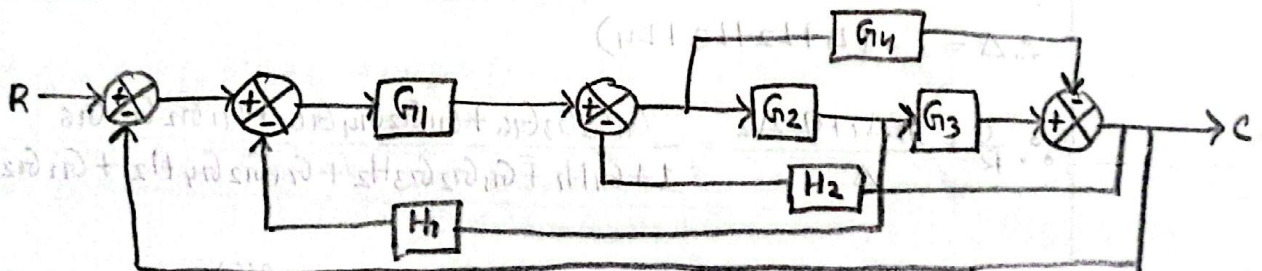
Two-non touching loops: None

$$\text{and } \Delta = 1 + G_1 G_2 - G_2 G_3$$

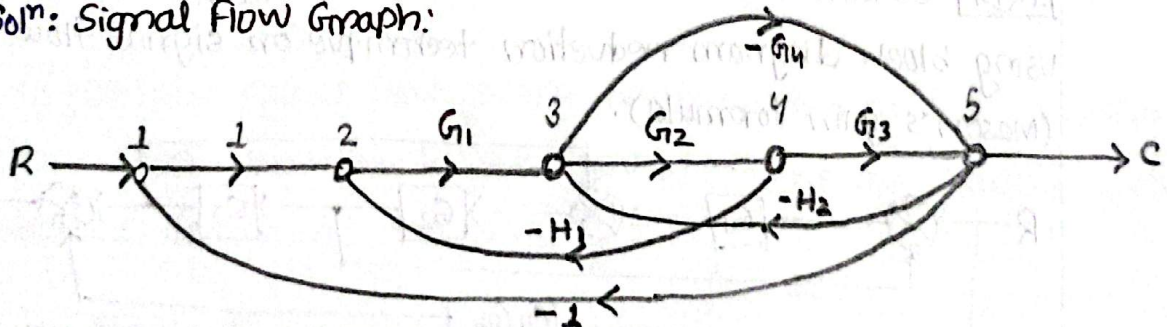
calculate Δ and Δ_k : $\Delta_1 = 1$, $\Delta_2 = 1 + G_1 G_2 - G_2 G_3$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 (1 + G_1 G_2 - G_2 G_3)}{1 + G_1 G_2 - G_2 G_3} \quad (\text{Ans})$$

1.45 Draw the signal flow graph and determine $\frac{C}{R}$ for the block diagram shown in the figure:



Soln: Signal Flow Graph:



The gain of forward paths: $g_1 = G_1 G_2 G_3$

$$g_2 = -G_1 G_4$$

Individual loops: $L_1 = -G_1 G_2 H_1$

$$L_2 = -G_2 G_3 H_2$$

$$L_3 = -G_1 G_2 G_3$$

$$L_4 = G_1 G_4$$

$$L_5 = G_4 H_2$$

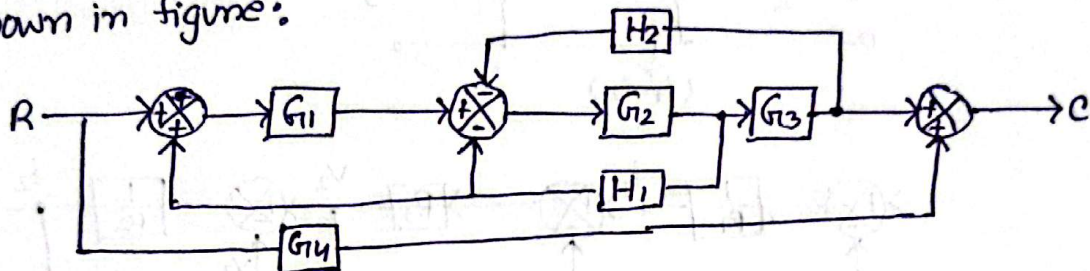
Two non-touching loops = none

calculate Δ and Δ_k : $\Delta_1 = 1$, $\Delta_2 = 1$

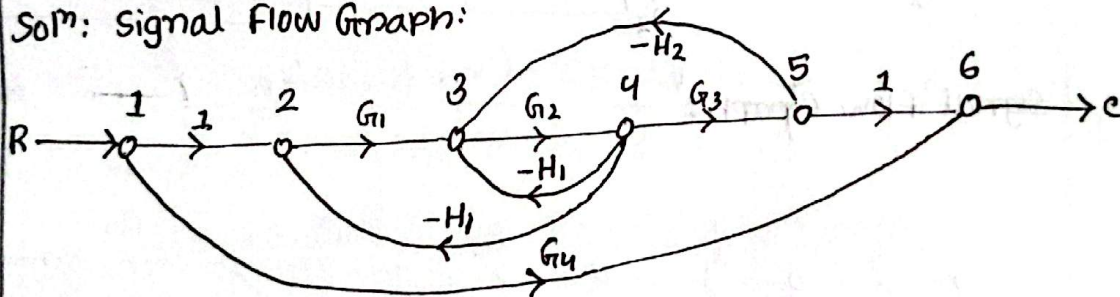
$$\therefore \Delta = 1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 - G_1 G_4 - G_4 H_2$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 - G_1 G_4}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 - G_1 G_4 - G_4 H_2} \quad (\text{Ans})$$

1.46 Obtain the transfer function $\frac{C}{R}$ of the block diagram shown in figure:



Soln: Signal Flow Graph:



The gain of the forward paths: $g_1 = G_1 G_2 G_3$
 $g_2 = G_4$

Individual Loops: $L_1 = -G_2 H_1$
 $L_2 = -G_1 G_2 H_1$
 $L_3 = -G_2 G_3 H_2$

Two-non touching loops: none

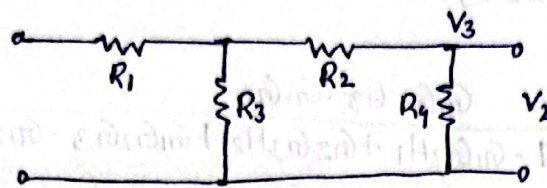
Calculating Δ and Δ_k : $\Delta_1 = 1$, $\Delta_2 = 1 - (-G_2 H_1 - G_1 G_2 H_1 - G_2 G_3 H_2)$
 $= 1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2$

$$\therefore \Delta = 1 - (L_1 + L_2 + L_3) = 1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2)}{1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2}$$

(Ans)

1.47 Draw the signal flow graph for the network shown in the figure, take V_3 as output node

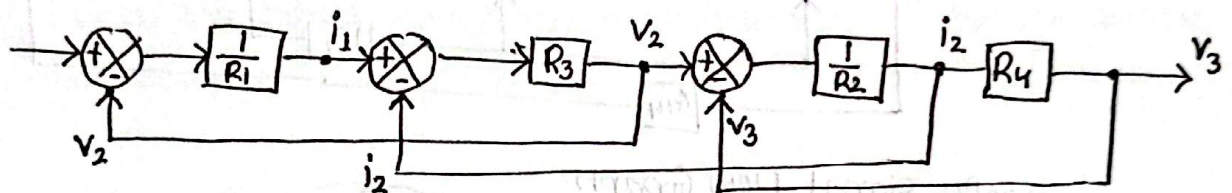
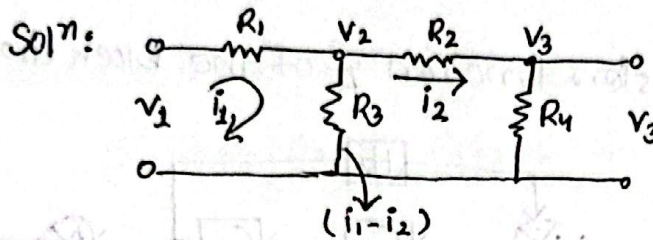


$$\text{Here, } i_1 = \frac{V_1 - V_2}{R_1}$$

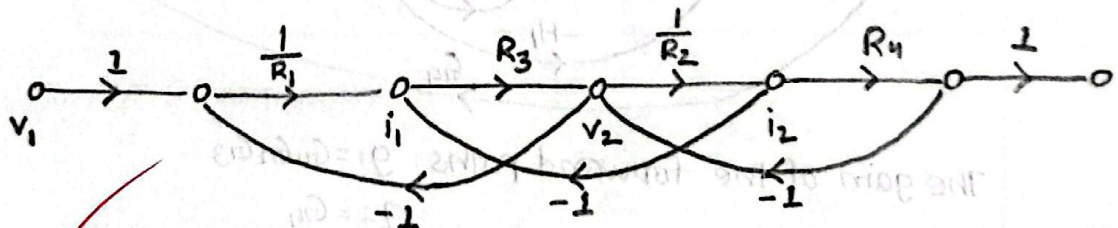
$$V_2 = (i_1 - i_2) R_3$$

$$i_2 = \frac{V_1 - V_3}{R_2}$$

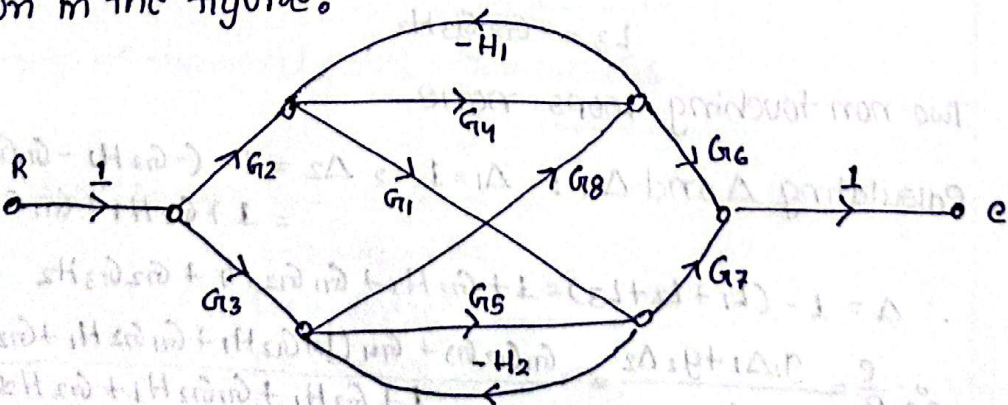
$$V_3 = i_2 R_4$$



Signal Flow Graph:



1.48 obtain the transfer function $\frac{C}{R}$ from the signal flow graph shown in the figure:



The gain of the forward paths: $g_1 = G_2 G_4 G_6$

$$g_2 = G_3 G_5 G_7$$

$$g_3 = G_2 G_1 G_7$$

$$g_4 = G_3 G_8 G_6$$

$$g_5 = -G_2 G_1 H_2 G_8 G_6$$

$$g_6 = -G_3 G_8 H_1 G_1 G_7$$

Individual Loops: $L_1 = -G_4 H_1$

$$L_2 = -G_5 H_2$$

$$L_3 = H_1 G_1 H_2 G_8$$

Two-non touching loops: $L_1 L_2 = G_4 H_1 \neq G_5 H_2$

calculate Δ and Δ_k : $\Delta_1 = 1 + G_5 H_2$, $\Delta_2 = 1 + G_4 H_1$,

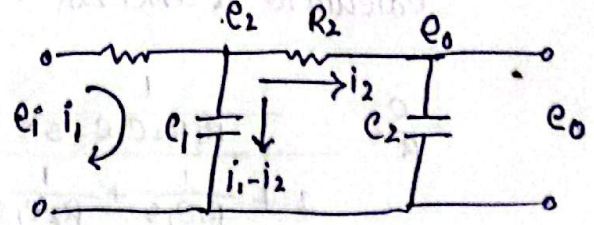
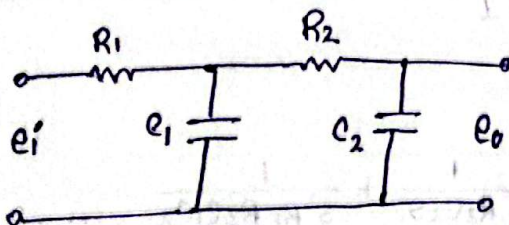
$$\Delta_3 = 1, \Delta_4 = 1, \Delta_5 = 1, \Delta_6 = 1$$

$$\therefore \Delta = 1 + G_4 H_1 + G_5 H_2 - H_1 G_1 H_2 G_8 + G_4 H_1 G_5 H_2$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2 + g_3 \Delta_3 + g_4 \Delta_4 + g_5 \Delta_5 + g_6 \Delta_6}{\Delta}$$

$$= \frac{G_2 G_4 G_6 (1 + G_5 H_2) + G_3 G_5 G_7 (1 + G_4 H_1) + G_1 G_2 G_7 + G_3 G_8 G_6 - G_2 G_1 H_2 G_8 G_6 - G_3 G_8 H_1 G_1 G_7}{1 + G_4 H_1 + G_5 H_2 - H_1 G_1 H_2 G_8 + G_4 H_1 G_5 H_2} \quad (\text{Ans})$$

1.49 Draw the block diagram, signal flow graph and find out the transfer function of the given circuit:



$$\text{Here, } i_1 = \frac{e_i - e_2}{R_1}, e_2 = \frac{1}{C_1} \int (i_1 - i_2) dt, i_2 = \frac{1}{R_2} (e_2 - e_o), e_o = \frac{1}{C_2} \int i_2 dt$$

Taking Laplace Transform:

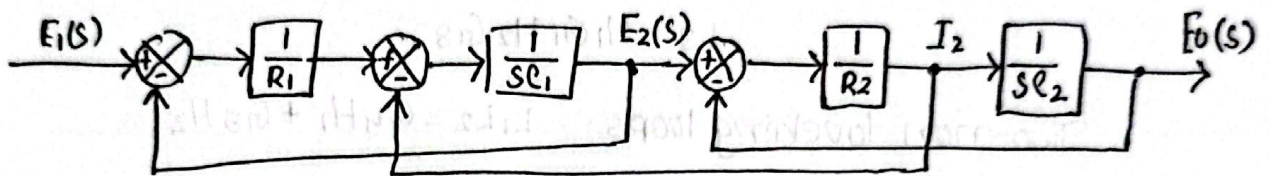
$$I_1(s) = \frac{1}{R_1} [E_1(s) - E_0(s)]$$

$$E_2(s) = \frac{1}{sC_1} [I_1(s) - I_2(s)]$$

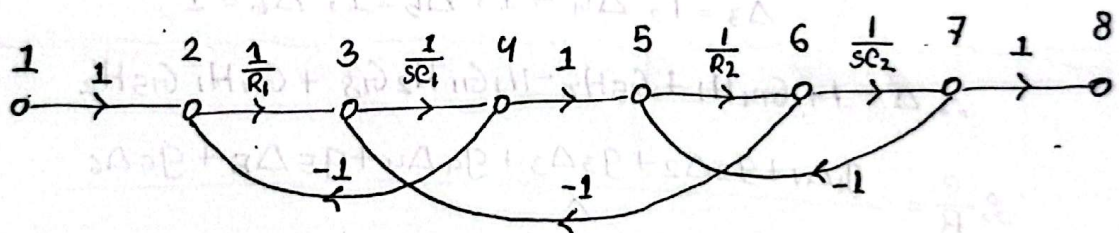
$$I_2(s) = \frac{1}{R_2} [E_2(s) - E_0(s)]$$

$$E_0(s) = \frac{1}{sC_2} [I_2(s)]$$

Block Diagram:



Signal Flow Graph:



The gain of forward path: $g_1 = \frac{1}{R_1 R_2 C_1 C_2 s^2}$

Individual Loops: $L_1 = -\frac{1}{R_1 C_1 s}$, $L_2 = -\frac{1}{R_2 C_1 s}$, $L_3 = -\frac{1}{R_2 C_2 s}$

Two non touching loops: $L_1 L_3 = \frac{1}{s^2 R_1 R_2 C_1 C_2}$

calculate Δ and Δ_k : $\Delta_1 = 1$

$$\frac{C}{R} = \frac{\frac{1}{R_1 R_2 C_1 C_2 s^2}}{1 + \frac{1}{R_1 C_1 s} + \frac{1}{R_2 C_1 s} + \frac{1}{R_2 C_2 s} + \frac{1}{s^2 R_1 R_2 C_1 C_2}}$$

$$= \frac{1}{1 + R_1 C_1 s + R_1 C_2 s + R_2 C_2 s + s^2 R_1 R_2 C_1 C_2}$$

(Ans)

05

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22.07.26

Assignment-02

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Section: 65-M

Chapter-03 (Error Analysis)

Example-3.1 The open loop transfer function of unity feedback system is given by $G(s) = \frac{50}{(1+0.1s)(s+10)}$

Determine the static error co-efficients k_p, k_v, k_a

Soln: Here, $G(s) = \frac{50}{(1+0.1s)(s+10)}$, unity feedback, so $H(s) = 1$

$$\text{We know that, } k_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{50 \times 1}{(1+0.1s)(s+10)} = \frac{50}{1 \times 10} = 5 \quad (\text{Ans})$$

$$k_v = \lim_{s \rightarrow 0} s G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{50}{(1+0.1s)(s+10)} = 0 \quad (\text{Ans})$$

$$k_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \cdot \frac{50}{(1+0.1s)(s+10)} = 0 \quad (\text{Ans})$$

Example-3.2 The forward path transfer function of a unity control system given by $G(s) = \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$. Determine the step, ramp and parabolic error co-efficients. Also determine the type of the system.

Soln: Here, $G(s) = \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$, $H(s) = 1$

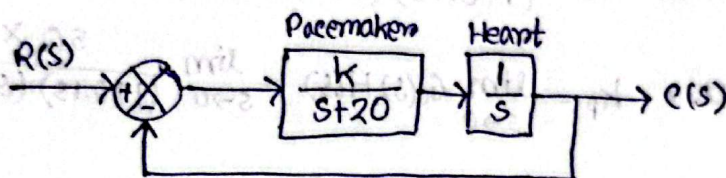
$$k_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)} = \infty \quad \therefore k_p = \infty \quad (\text{Ans})$$

$$k_v = \lim_{s \rightarrow 0} s G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)} = \infty \quad \therefore k_v = \infty \quad (\text{Ans})$$

$$k_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \cdot \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)} = \frac{10}{5} = 2 \quad \therefore k_a = 2 \quad (\text{Ans})$$

In denominator the value of $m=2$. Hence, the given system is type '2' system.

Example-3.3 The block diagram of an electronic pacemaker is given in figure. Determine the steady state error for unit ramp input when $k=400$. Also determine the value of k for which the steady state error to a unit ramp will be 0.02



Solⁿ: Given that, $k=400$, $R(s) = \frac{1}{s^2}$, $H(s) = 1$, $G(s) = \frac{k}{s(s+20)}$

$$\therefore G(s)H(s) = \frac{k}{s(s+20)}$$

$\therefore$ Steady state error is given by, $e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$

$$= \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{k}{s(s+20)}} = \lim_{s \rightarrow 0} \frac{s+20}{s(s+20)+400} = \frac{20}{400} = 0.05 \text{ (Ans)}$$

Now, Given that, $e_{ss} = 0.02$

$$\text{So, } e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{k}{s(s+20)}} \Rightarrow 0.02 = \lim_{s \rightarrow 0} \frac{s+20}{s(s+20)+k}$$

$$\Rightarrow 0.02 = \frac{20}{k} \quad \therefore k=1000 \quad \text{(Ans)}$$

(3)

Example - 3.4 For a unity control system the forward path transfer function is given by $G(s) = \frac{20}{s(s+20)(s^2+2s+20)}$. Determine the steady state error of the system when the inputs are (i) 5 (ii) 5t (iii) $\frac{3t^2}{2}$

Soln: (i) Given that, $r(t) = 5 \therefore R(s) = \frac{5}{s}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1 + G(s)H(s)} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s} \cdot \frac{1}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{5s(s+2)(s^2+2s+20)}{(s+2)(s^2+2s+20)+20} = \frac{0}{20} = 0 \quad \therefore e_{ss} = 0 \quad (\text{Ans})$$

(ii) Here, $R(s) = \frac{5}{s^2}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s^2} \cdot \frac{1}{1 + \frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s^2} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$= \frac{200}{20} = 10 \quad \therefore e_{ss} = 10 \quad (\text{Ans})$$

(iii) Here, $R(s) = \frac{3}{s^3}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{3}{s^3} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20} = \infty \quad \therefore e_{ss} = \infty \quad (\text{Ans})$$

Example - 3.5 The open loop transfer function of a unity feedback system is given by $G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$. Find the static error co-efficients and the steady state error of the system when subjected to an input given by $r(t) = 2 + 5t + 2t^2$

Soln: Here, $G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$, $r(t) = 2 + 5t + 2t^2$, $H(s) = 1$

$$\therefore k_p = \lim_{s \rightarrow 0} G(s) \cdot H(s) = \lim_{s \rightarrow 0} \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$\therefore k_v = \lim_{s \rightarrow 0} s \cdot G(s) \cdot H(s) = \lim_{s \rightarrow 0} s \cdot \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$\therefore k_a = \lim_{s \rightarrow 0} s^2 \cdot G(s) \cdot H(s) = \lim_{s \rightarrow 0} \frac{108}{s(s+4)(s^2+3s+12)} = \frac{108}{48}$$

$$\text{Now, } r(t) = 2 + 5t + 2t^2 \quad \therefore R(s) = \frac{2}{s} + \frac{5}{s^2} + \frac{4}{s^3}$$

$$\therefore e_{ss} = \frac{R_1}{1+k_p} + \frac{R_2}{k_v} + \frac{R_3}{k_a} = \frac{2}{1+\infty} + \frac{5}{\infty} + \frac{4 \times 48}{108} = 1.77$$

$$\therefore e_{ss} = 1.77 \quad (\text{Ans})$$

Example-3.6 Determine the type of the following unity feedback systems for which the forward path transfer function are given

(a) $G(s) = \frac{k}{(s+10)(s+5)}$ (b) $G(s) = \frac{20}{s(1+0.1s)(1+0.05s)}$

(c) $G(s) = \frac{20(s+2)}{s^2(s+2)(s^2+5s+5)}$ (d) $G(s) = \frac{k}{s(s+1)(s+2)}$

Soln:

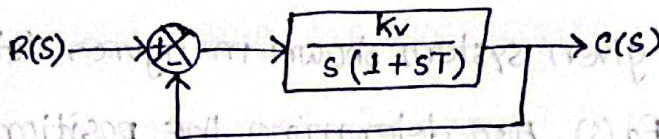
(a) Since in denominator the power of s is zero so, $m=0$;
hence it is type zero system. (Ans)

(b) Since in denominator the power of s is one so, $m=1$;
hence it is type one system. (Ans)

(c) Since in denominator the power of s is three so, $m=3$;
hence it is type '3' system. (Ans)

(d) Since in denominator the power of s is one so, $m=1$;
hence it is type '1' system. (Ans)

3.7th Example A servomechanism is designed to keep a radar antenna pointed at a flying aeroplane. If the aeroplane is flying with a velocity of 600 km/hr, at a range of 2 km, and the maximum tracking error is to be within 0.1° , determine the required velocity and error co-efficient



Soln: Here, Linear velocity = 600 km/h, $r = 2$ km

$$\therefore \text{Angular velocity, } \omega = \frac{600}{2} = 300 \text{ rad/hr.} = \frac{300}{3600} \text{ rad/sec} = 0.083 \text{ rad/sec}$$

$$\therefore R(s) = \frac{0.083}{s^2}$$

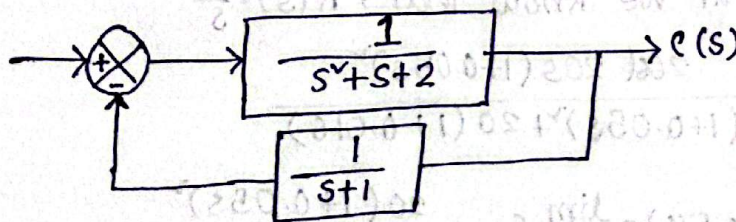
$$\text{Now, } e_{ss} = \lim_{s \rightarrow 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{0.083}{s^2} \cdot \frac{1}{1 + \frac{K_v}{s(1+sT)}}$$

$$\Rightarrow e_{ss} = \lim_{s \rightarrow 0} \frac{0.083}{s} \cdot \frac{s(1+sT)}{s(1+sT) + K_v} \Rightarrow 0.1 = \frac{0.083}{K_v}$$

$$\therefore K_v = 0.083 \text{ per degree}$$

$\therefore$ Required Velocity Error Co-efficient, $K_v = 0.083$ per degree (Ans)

Example 3.8 for the system shown in Figure, Determine K_p and e_{ss} for unit step input.

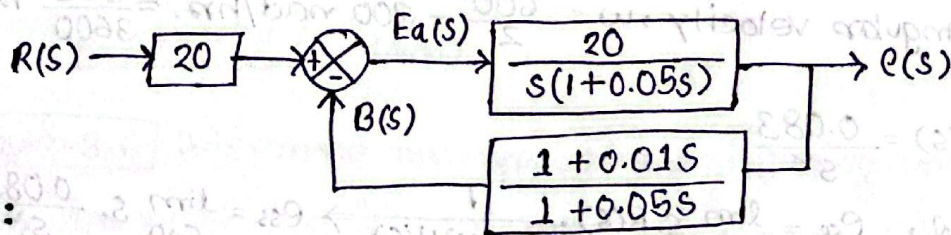


Soln: Here, $G(s) = \frac{1}{s^2 + s + 2}$, $H(s) = \frac{1}{s+1}$

$$\therefore k_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{1}{(s^2 + s + 2)(s+1)} = \frac{1}{2} \quad \therefore k_p = \frac{1}{2} \quad (\text{Ans})$$

$$\therefore e_{ss} = \frac{1}{1 + k_p} = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3} \quad \therefore e_{ss} = \frac{2}{3} \quad (\text{Ans})$$

Example 3.9 For a given system shown in Figure, determine the actuating signal $E_a(s)$. Also determine the position error constant for unit step input.



Soln:

Here, $E_a(s) = 20 R(s) - B(s)$, $C(s) = \frac{20 E_a(s)}{s(1+0.05s)}$

$$B(s) = C(s) \cdot \frac{1+0.01s}{1+0.05s} \Rightarrow B(s) = \frac{20(1+0.01s) E_a(s)}{s(1+0.05s)^2}$$

$$\therefore E_a(s) = 20 R(s) - \frac{20(1+0.01s) E_a(s)}{s(1+0.05s)^2} \Rightarrow E_a(s) \left[1 + \frac{20(1+0.01s)}{s(1+0.05s)^2} \right] = 20 R(s)$$

$$\Rightarrow E_a(s) = \frac{20 R(s)}{1 + \frac{20(1+0.01s)}{s(1+0.05s)^2}}$$

$$\therefore E_a(s) = \frac{20s(1+0.05s)^2 R(s)}{s(1+0.05s)^2 + 20(1+0.01s)} \quad (\text{Ans})$$

For step input we know that, $R(s) = \frac{1}{s}$

$$E_a(s) = \frac{20s(1+0.05s)^2}{s(1+0.05s)^2 + 20(1+0.01s)}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot \frac{20(1+0.05s)^2}{s(1+0.05s)^2 + 20(1+0.01s)} = 0$$

But, $e_{ss} = \frac{1}{1+k_p} \Rightarrow 0 = \frac{1}{1+k_p} \quad \boxed{\therefore k_p = \infty} \quad (\text{Ans})$

Example 3.10 Consider a unity feedback control system with the closed loop transfer function $\frac{C(s)}{R(s)} = \frac{ks+b}{s^2+as+b}$, determine the open loop transfer function. Show that the steady state error in the unit ramp input response is given by $e_{ss} = \frac{a-k}{b}$

Solⁿ: we know that, for closed loop control system:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}, \quad H(s)=1 \quad \therefore \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$

Given that, $\frac{C(s)}{R(s)} = \frac{ks+b}{s^2+as+b} \quad \therefore \frac{ks+b}{s^2+as+b} = \frac{G(s)}{1+G(s)}$

$$\therefore G(s) = \frac{ks+b}{s^2+s(a-k)}$$

$\therefore$ Open loop transfer function $G(s) = \frac{ks+b}{s^2+s(a-k)} \quad (\text{Ans})$

$\therefore$ Steady State Error, $e_{ss} = \lim_{s \rightarrow 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)}$

$$= \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{ks+b}{s^2+s(a-k)} \cdot 1} = \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s^2+s(a-k)}{s^2+s(a-k)+ks+b}$$

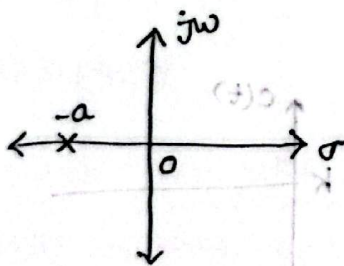
$$= \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s[s+a-k]}{s^2+s(a-k)+ks+b} = \frac{a-k}{b}$$

$\therefore e_{ss} = \frac{a-k}{b} \quad (\text{proved})$

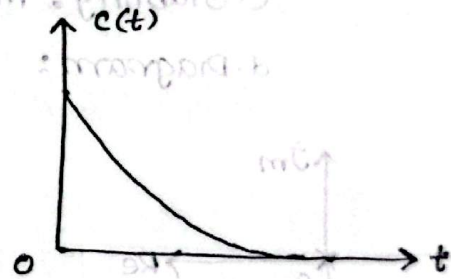
Assignment - 03Stability TheoryEffect of Location of poles on stability:① Simple Pole on Negative Real Axisa. Pole Position: $s = -a$ b. Impulse Response: $g(t) = \mathcal{L}^{-1} \frac{k}{s+a} = k \cdot e^{-at}$ c. Behavior: The response exponentially decays to zero as t increases.

d. Stability: stable

e. Diagram:



(a) Simple pole on negative real axis (s-plane)

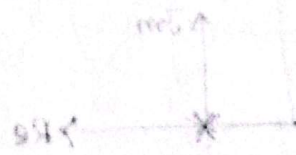
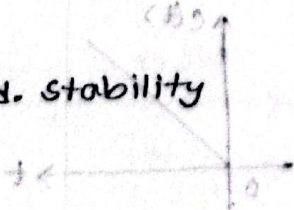


(b) Time Response

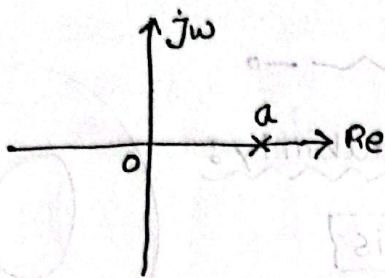
② Simple Pole on Positive Real Axisa. Pole Position: $s = +a$ b. Impulse Response: $g(t) = \mathcal{L}^{-1} \frac{k}{s-a} = k e^{at}$

c. Behavior: The response increases exponentially with time

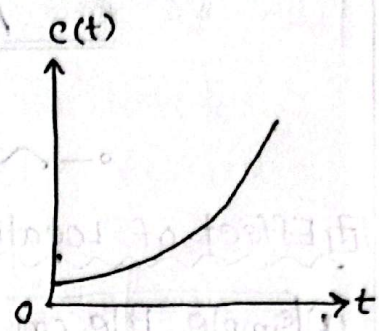
d. Stability: unstable

P.T.O

e. diagram:



(a) Simple pole on positive real axis



(b) time response

③ Pole at Origin

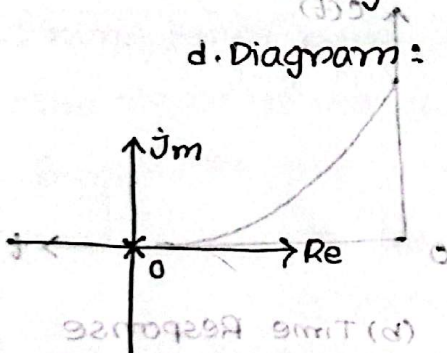
(i) Single Pole at Origin ($s=0$):

a. Response: $g(t) = L^{-1} \frac{k}{s} = k$

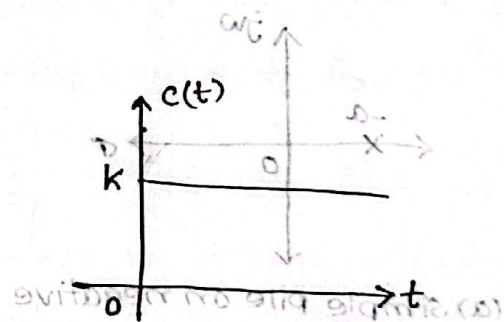
b. Behavior: constant value over time

c. Stability: Marginally stable

d. Diagram:



(a) Single pole at origin



(b) Response due to single pole

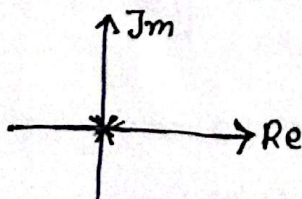
(ii) Double Pole at Origin ($s^2=0$):

a. Response: $g(t) = L^{-1} \frac{k}{s^2} = kt$

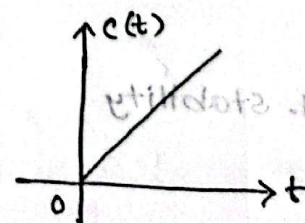
b. Behavior: Increases linearly with time.

c. Stability: Unstable

d. Diagram:



(a) Double Pole at Origin



(b) Impulse Response

④ Complex Conjugate Poles in Left Half of s-plane

a. Pole Position: $s = -\alpha \pm j\omega$

b. Impulse Response: $g(t) = 2ke^{-\alpha t} \cos(\omega t)$

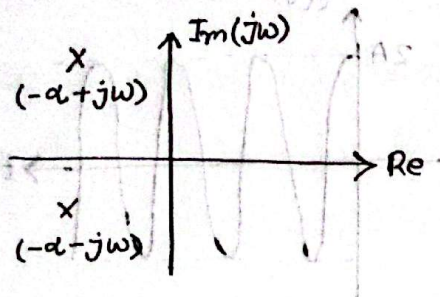
Here,

$$g(t) = \mathcal{L}^{-1} \left[\frac{k}{s + \alpha - j\omega} + \frac{k}{s + \alpha + j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2k(s + \alpha)}{(s + \alpha)^2 + \omega^2} \right] = 2ke^{-\alpha t} \cos \omega t$$

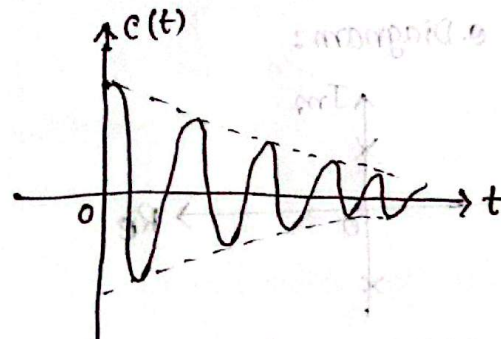
c. Behavior: Damped Oscillations that eventually decay to zero.

d. Stability: Stable

e. Diagram:



(a) s-plane



(b) Time Response

⑤ Complex Conjugate Poles in Right Half of s-plane

a. Pole Position: $s = +\alpha \pm j\omega$

b. Impulse Response: $g(t) = 2Ae^{\alpha t} \cos(\omega t)$

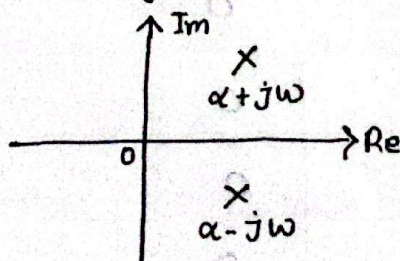
Here,

$$g(t) = \mathcal{L}^{-1} \left[\frac{A}{s - \alpha - j\omega} + \frac{A}{s - \alpha + j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2A(s - \alpha)}{(s - \alpha)^2 + \omega^2} \right] = 2Ae^{\alpha t} \cos \omega t$$

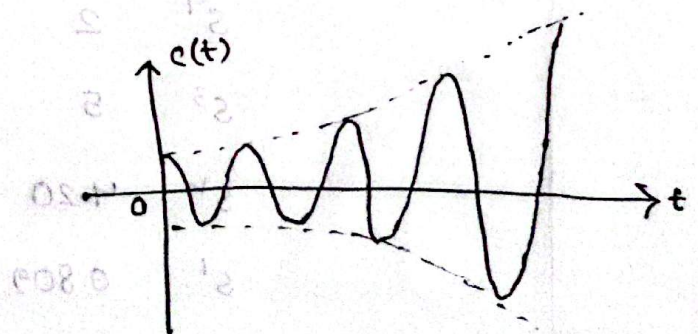
c. Behavior: Oscillations grow exponentially over time

d. Stability: Unstable

e. Diagram:



(a) s-plane



(b) Time Response

(4)

⑥ Complex Conjugate Poles on $j\omega$ -axis (Imaginary Axis)

a. Pole Position: $s = \pm j\omega$

b. Impulse Response: $g(t) = 2A \cos(\omega t)$

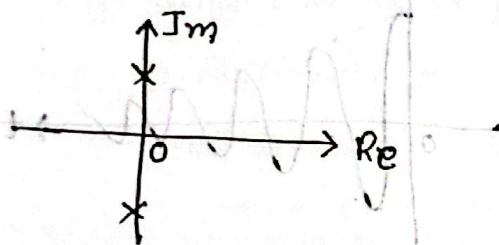
c. Behavior: Here,

$$g(t) = \mathcal{L}^{-1} \left[\frac{A}{s+j\omega} + \frac{A}{s-j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2As}{s^2 + \omega^2} \right] = 2A \cos \omega t$$

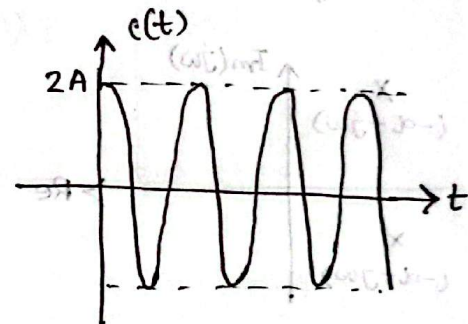
d. Behavior: Sustained oscillations of constant amplitude

e. Stability: Marginally stable (or unstable depending on context)

f. Diagram:



(a) Poles on Imaginary Axis



(b) Time Response

Routh Stability

Example 5.3 Determine the stability of the system having following

characteristics equation: $2s^4 + 5s^3 + 5s^2 + 2s + 1 = 0$

Soln: Here, $2s^4 + 5s^3 + 5s^2 + 2s + 1 = 0$

s^4	2
s^3	5
s^2	4.20
s^1	0.809
s^0	1

5	1
2	0
1	0
0	0
0	0

5

From the Routh table:

No. of sign changes in first column = 0

No. of poles on the right hand side of s-plane = 0

Hence, the system is stable. (Ans)

Example 5.8 Apply Routh-Hurwitz criterion to the following equation and investigate the stability

$$s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0$$

Soln: Here,

$$\text{Sign } s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0$$

$$+ s^5 \quad 1 \quad 2 \quad 11$$

$$+ s^4 \quad 2 \quad 4 \quad 10$$

$$+ s^3 \quad 0 \quad 6 \quad 0$$

$$- s^2 \quad \frac{4s-12}{s} \quad 10 \quad 0$$

$$+ s^1 \quad \frac{24s-72-10s^2}{s} \quad 0 \quad 0$$

$$+ s^0 \quad 10 \quad 0 \quad 0$$

Here,

No. of sign change = 2

No. of poles at R.H.P = 2

∴ Hence, the system is unstable

(Ans)

calculations

$$(2 \times 2 - 4 \times 1) / 0 = 0$$

$$(2 \times 11 - 1 \times 10) / 2 = 6$$

$$\frac{4s-12}{s} = \frac{4s-12}{s}$$

$$\frac{10s}{s} = 10$$

$$\frac{24s-72}{6} - 10s = \frac{24s-72-10s^2}{s}$$

$$\frac{10 \times \left(\frac{24s-72-10s^2}{s} \right) - 0 \times \left(\frac{4s-12}{s} \right)}{\frac{24s-72-10s^2}{s}} = 10$$

$$\lim_{s \rightarrow 0} \frac{4s-12}{s} = \lim_{s \rightarrow 0} \frac{4s}{s} - \frac{12}{s} = 4 - \frac{12}{0} = -\infty$$

$$\lim_{s \rightarrow 0} \frac{24s}{s} - \frac{72}{s} + \frac{10s^2}{s} = \frac{-72}{-12} = +6$$

$$0 = (s+2)(s-2)(s+3)(s+1)(s+1)$$

P.T.O

⑥

Example 5.10 Consider the equation: $s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$

Soln: Here, $s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$

sign					calculation
+	s^5	1	24	-25	$(2 \times 24 - 48 \times 1) / 2 = 0$
+	s^4	2	48	-50	$(2 \times (-25) - 1 \times (-50)) / 2 = 0$
+	s^3	0 ⁸	0 ⁹⁶	0	Here, $A(s) = 2s^4 + 48s^2 - 50$ $\frac{d}{ds} A(s) = 8s^3 + 96s$
+	s^2	24	-50	0	$(48 \times 8 - 96 \times 2) / 8 = 24$
+	s^1	112.6	0	0	$\{8 \times (-50) - 0\} / 8 = -50$
-	s^0	-50	0	0	$\{24 \times 96 - 8 \times (-50)\} / 24 = 112.6$ $\{112.6 \times (-50) - 24 \times 0\} / 112.6 = -50$

∴ number of sign change = 1

∴ number of poles at right hand side of s-plane = 1

∴ The roots of the equation can be written in factored form as formed by the auxiliary polynomial $A(s) = 0$

⇒ Then, $s^2 = x$

$$\therefore 2x^2 + 48x - 50 = 0$$

$$\Rightarrow x^2 + 50x - 2x - 50 = 0 \Rightarrow x^2 + 24x - 25 = 0$$

$$\Rightarrow x(x + 50) - 2x^2 + 25x - x - 25 = 0$$

$$\Rightarrow (x - 1)(x + 25) = 0$$

Now,

$$x = 1$$

$$\Rightarrow s^2 = 1$$

$$\Rightarrow s = \pm 1$$

Again,

$$x = -25$$

$$\Rightarrow s^2 = -25$$

$$\Rightarrow s^2 = j^2 5^2$$

$$\Rightarrow s = \pm j5$$

} so two imaginary poles

∴ The given equation can be written in factored form as

$$(s + 1)(s - 1)(s + j5)(s - j5)(s + 2) = 0$$